Project Iteration 3: Testing and Initial Development

Introduction

Project Name: Pages & Prose

Team Number: 8

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Course: DIGT2107 Practice of Software Development

Requirement to Test Case Traceability

Requirement ID	Requirement Description	Test Case ID	Test Case Description	Coverage
FR1	Account registration, sign-in and sign-out with session ending. Use case: UC-11	TC-001 TC-031 TC-038	TC-001: Account registration and session lifecycle TC-031: Check real sign-in and access rules TC-038: Guest access boundary	Uncovered
FR2	View public profiles and edit own display name and description. Use case: UC-11	TC-002 TC-031 TC-032 TC-040	TC-002: Edit profile and retain community reviews TC-031: Check real sign-in and access rules TC-032: Keep saved data when the app reopens TC-040: Public profile review and discussion visibility	Partially Covered
FR3	Search title/author and view book details, including a clear empty result. Use case: UC-01	TC-003 TC-004 TC-039 TC-042	TC-003: Search title or author and read details TC-004: Empty and unmatched search TC-039: Browser reading and discussion journey TC-042: Reject unknown resources and unshelved progress	Fully Covered

Requirement ID	Requirement Description	Test Case ID	Test Case Description	Coverage
FR4	Keep one main reading status per reader and book. Add, move, and remove books; set or clear completion dates; support custom shelves. Use case: UC-02	TC-005 TC-006 TC-007 TC-008 TC-012 TC-031 TC-039 TC-043 TC-044	TC-005: Add and move one primary shelf record TC-006: Completion date lifecycle TC-007: Remove only the current reader shelf entry TC-008: Custom shelf organisation TC-012: Prompt before recording completion TC-031: Check real sign-in and access rules TC-039: Browser reading and discussion journey TC-043: Demo reading-list separation TC-044: Completion date remains stable on repeat saves	Partially Covered
FR5	Accept whole pages 0 to known positive total; calculate page/total x 100; disable unknown totals and prompt for completion at 100%. Use case: UC-03	TC-009 TC-010 TC-011 TC-012 TC-039 TC-042 TC-043	TC-009: Progress boundaries and calculation TC-010: Reject invalid progress without data loss TC-011: Unknown and zero page totals TC-012: Prompt before recording completion TC-039: Browser reading and discussion journey TC-042: Reject unknown resources and unshelved progress TC-043: Demo reading-list separation	Fully Covered
FR6	Publish, browse, edit and delete own ratings/reviews; show review on book and profile.	TC-013 TC-014 TC-015 TC-031 TC-037	TC-013: Review create edit delete and profile link TC-014: Quarter-star boundaries and increments	Partially Covered

Requirement ID	Requirement Description	Test Case ID	Test Case Description	Coverage
	Use cases: UC-04, UC-05	TC-039 TC-040 TC-042	TC-015: Only edit or delete your own review TC-031: Check real sign-in and access rules TC-037: Render untrusted review text safely TC-039: Browser reading and discussion journey TC-040: Public profile review and discussion visibility TC-042: Reject unknown resources and unshelved progress	
FR7	Book-specific threads/replies with contributor display name and posting time. Use case: UC-07	TC-016 TC-017 TC-039 TC-040 TC-042	TC-016: Book-specific threads and replies TC-017: Reject invalid discussion contributions TC-039: Browser reading and discussion journey TC-040: Public profile review and discussion visibility TC-042: Reject unknown resources and unshelved progress	Fully Covered
FR8	Contributors flag spoiler content; hide text until deliberate reveal. Page/percentage threshold extension remains in US8 backlog. Use case: UC-07	TC-018 TC-019 TC-039	TC-018: Spoilers hidden until explicit reveal TC-019: Page-based spoiler thresholds TC-039: Browser reading and discussion journey	Fully Covered
FR9	Read supported EPUBs; owner-only passage notes with create/revisit/delete;	TC-020 TC-031 TC-036	TC-020: EPUB notes privacy and invalid files TC-031: Check real sign-in and access rules	Uncovered

Requirement ID	Requirement Description	Test Case ID	Test Case Description	Coverage
	safe errors for unreadable files. Use case: UC-12		TC-036: Private records cannot be retrieved by another account	
FR10	Private date-filtered summaries of completed books, grouped by genre and author. Use case: UC-09	TC-021 TC-036	TC-021: Date-filtered reading summaries TC-036: Private records cannot be retrieved by another account	Uncovered
FR11	Quarter-star overall rating; optional prose/characters/plot sub-ratings and content warnings; preserve old reviews. Use cases: UC-04, UC-05	TC-014 TC-022	TC-014: Quarter-star boundaries and increments TC-022: Optional sub-ratings warnings and migration	Partially Covered
FR12	Book-specific buddy groups with invitations, accept/reject and accepted-member-only discussion. Use case: UC-13	TC-023 TC-036	TC-023: Buddy-read invitation and membership privacy TC-036: Private records cannot be retrieved by another account	Uncovered
FR13	Suggest books with the most reviews first. If review counts tie, sort by title. Each suggestion opens its book details. Use case: UC-14	TC-024	TC-024: Deterministic community recommendations	Uncovered
FR14	Allow author applications for specific books; admin approves verification and labelled author participation. Use case: UC-15	TC-025	TC-025: Verified author application and admin decision	Uncovered

Requirement ID	Requirement Description	Test Case ID	Test Case Description	Coverage
FR15	Follow/unfollow readers; suggestions and feed expose only community-visible activity. Use case: UC-06	TC-026	TC-026: Follow feed without private activity leakage	Uncovered
FR16	Readers can report content. Only admins can review reports, dismiss them, remove content, or issue warnings. Each decision is recorded. Use cases: UC-08, UC-10	TC-027	TC-027: Report and moderate with role enforcement	Uncovered

Coverage summary: 4 requirements are fully tested (FR3, FR5, FR7, FR8), 4 are partly tested (FR2, FR4, FR6, FR11), and 8 are not yet tested (FR1, FR9, FR10, FR12-FR16). All 16 have planned test cases. These counts describe requirement testing, not how many lines of code were tested.

Non Functional Test Coverage

Requirement ID	Requirement Description	Test Case ID	Test Case Description	Coverage
NFR1	With 100 sample books and one active user, at least 19/20 searches and 19/20 shelf updates finish within 2 seconds from submission to displayed result.	TC-028	TC-028: Measure 100-book browser performance	Not required
NFR2	Add from details in at most three actions; at least 4/5 first-time readers complete search, add and status change within 3 minutes with only a task prompt.	TC-029	TC-029: First-time usability and action count	Not required
NFR3	Core controls have visible labels or accessible names, keyboard operation and visible focus; core	TC-030	TC-030: Keyboard access labels and focus sample	Not required

Requirement ID	Requirement Description	Test Case ID	Test Case Description	Coverage
	workflow is mouse-free; status/errors do not rely on colour alone.			
NFR4	Authenticated operations reject changes to another user profile, shelf, review or note; dedicated password hashing; sign-out invalidates session.	TC-001 TC-015 TC-020 TC-025 TC-027 TC-031 TC-037	TC-001: Account registration and session lifecycle TC-015: Only edit or delete your own review TC-020: EPUB notes privacy and invalid files TC-025: Verified author application and admin decision TC-027: Report and moderate with role enforcement TC-031: Check real sign-in and access rules TC-037: Render untrusted review text safely	Not required
NFR5	Saved personal records survive reopening; failed save shows error, retains last saved data and never claims success. In-memory adapter is test-only.	TC-010 TC-022 TC-032 TC-033 TC-039 TC-041	TC-010: Reject invalid progress without data loss TC-022: Optional sub-ratings warnings and migration TC-032: Keep saved data when the app reopens TC-033: Failed saves keep existing data TC-039: Browser reading and discussion journey TC-041: Browser save error and truthful feedback	Not required
NFR6	Separate presentation, application rules and storage. Adapter replacement requires no changes to search or shelf screens; verify same checks and dependencies.	TC-034	TC-034: Change storage without changing the screens	Not required
NFR7	Future persistent service: 10,000 books, 50 concurrent users, five-minute trial; at least 95% of searches/updates within 3 seconds and error rate below 1%;	TC-035	TC-035: Future test with many books and users	Not required

Requirement ID	Requirement Description	Test Case ID	Test Case Description	Coverage
	record hardware/workload.			
NFR8	Shelves, progress, EPUBs, notes and summaries are owner-private; profiles/reviews/public forum community-visible; groups accepted-member-only; verify second-account retrieval.	TC-007 TC-020 TC-021 TC-023 TC-026 TC-036 TC-040 TC-043	TC-007: Remove only the current reader shelf entry TC-020: EPUB notes privacy and invalid files TC-021: Date-filtered reading summaries TC-023: Buddy-read invitation and membership privacy TC-026: Follow feed without private activity leakage TC-036: Private records cannot be retrieved by another account TC-040: Public profile review and discussion visibility TC-043: Demo reading-list separation	Not required

Next steps: add real sign-in and access checks; finish custom shelves and extra review fields; test the app with new readers; finish the keyboard checks; and build shared data storage before testing many users or private records. Tests that cannot run yet are marked Blocked.

Test Plan and Actual Results

Test data: We use Alice and Bob as demo readers. Book b1 has 432 pages, b2 has 280 pages, and b5 has no page total. Unit tests use a fixed date. Browser tests start with empty storage. The speed test uses 100 sample books. Future privacy tests will need a third user, real sign-in, and group invitations.

Test environment: Windows 11 (10.0.26300), Node 22.21.0, Chrome 157.0.8103.38, and Playwright. The computer has an AMD Ryzen 7 8700F processor and 32 GB of RAM. The browser window is 1440 x 1050 pixels. The app runs locally and saves data in the browser. The updated prototype passed all 24 unit tests and five browser checks on November 7, 2026 at 6:48 p.m. Toronto time.

Result category	Recorded outcome
Designed cases	44 unique cases covering 16 FRs and eight NFRs.
Unit suite	24 tests passed, 0 failed; node --test tests/core.test.mjs; raw TAP in docs/testing/unit-results.tap.

Result category	Recorded outcome
Browser suite	5 checks passed, 0 failed; tests/browser.mjs; raw environment/timings in docs/testing/browser-results.json.
Case status totals	29 Pass and 15 Blocked. A blocked test needs a feature or test setup that is not available yet.
Regression discovered	The first keyboard search test lost focus after the screen refreshed. The code now returns focus to Search, and the browser tests passed when rerun.
Exit assessment	The available tests passed for the local prototype. Real account security, full privacy, new-reader testing, accessibility, and support for many users still need work.

Execution and Defect Handling

Start each test with fresh sample data. Follow the steps, compare the result with what was expected, and save the output or screenshots. Use Pass when the test meets its expected result, Fail when it does not, and Blocked when a needed feature or setup is missing. For each problem, record the test ID, steps, expected and actual results, severity, and person responsible. After a fix, rerun the affected tests. If the fix changes shared data, rerun all unit tests and the full browser workflow.

Detailed Test Cases

Field	Details
Test Case ID	**TC-001**
Title	Account registration and session lifecycle
Related Requirement ID(s)	FR1, NFR4
Preconditions	Future authentication service; unused email; two browser sessions.
Steps	1. Register a reader with a valid email and password. 2. Reject duplicate email and wrong password. 3. Sign in, capture the session, then sign out. 4. Retry a protected request with the old session.
Expected Result	Registration succeeds once; passwords are hashed with a dedicated password hash; invalid credentials fail; sign-out invalidates the session.

Field	Details
Actual Result	Blocked: the demo has no sign-up, password storage, or real sign-in sessions.
Status	Blocked
Priority	High
Test Case ID	**TC-002**
Title	Edit profile and retain community reviews
Related Requirement ID(s)	FR2
Preconditions	Alice selected; fresh service and b1 available.
Steps	1. Publish a 4-star b1 review. 2. Save display name " Alice A " and bio " Bio ". 3. Read Alice profile and its reviews. 4. Attempt a blank display name.
Expected Result	Name and bio are trimmed and updated; review remains visible; blank name is rejected.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-003**
Title	Search title or author and read details
Related Requirement ID(s)	FR3
Preconditions	Fresh catalogue with b1 Pride and Prejudice and b2 Frankenstein.
Steps	1. Search " PRIDE ". 2. Search "shelley". 3. Open b1 details.
Expected Result	Queries are case-insensitive and trimmed; correct book returned; details show title, author and 432 pages.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass

Field	Details
Priority	High
Test Case ID	**TC-004**
Title	Empty and unmatched search
Related Requirement ID(s)	FR3
Preconditions	Five-book sample catalogue.
Steps	1. Submit whitespace only. 2. Search "zzzzzz".
Expected Result	Whitespace returns all five books; unmatched query returns an empty result set; UI no-results text is checked during manual follow-up.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-005**
Title	Add and move one primary shelf record
Related Requirement ID(s)	FR4
Preconditions	Alice selected and b1 absent from her shelves.
Steps	1. Add b1 as Want to Read. 2. Move b1 to Currently Reading. 3. Attempt Favourite Read as main status.
Expected Result	Exactly one record remains with Currently Reading; invalid main reading status is rejected.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-006**
Title	Completion date lifecycle

Field	Details
Related Requirement ID(s)	FR4
Preconditions	Alice selected; fixed test clock 2026-11-07T18:00:00Z.
Steps	1. Mark b1 Completed. 2. Inspect completedAt. 3. Move b1 to Want to Read.
Expected Result	Completion date equals the test clock; leaving Completed clears that date.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-007**
Title	Remove only the current reader shelf entry
Related Requirement ID(s)	FR4, NFR8
Preconditions	Alice and Bob each have b1 on a shelf.
Steps	1. As Bob remove b1. 2. Read Bob shelves. 3. Switch to Alice and read her shelves.
Expected Result	Bob has no b1 entry; Alice entry is retained.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-008**
Title	Custom shelf organisation
Related Requirement ID(s)	FR4
Preconditions	Future custom-shelf UI and service; Alice owns a b1 entry.

Field	Details
Steps	1. Create shelf "Summer". 2. Assign b1 while retaining its main status. 3. Rename shelf then remove the custom membership.
Expected Result	Custom label changes do not duplicate or erase the main status or progress.
Actual Result	Blocked: custom shelf creation and membership are not implemented.
Status	Blocked
Priority	Medium
Test Case ID	**TC-009**
Title	Progress boundaries and calculation
Related Requirement ID(s)	FR5
Preconditions	Alice has b1 Currently Reading; total 432 pages.
Steps	1. Save pages 0, 216 and 432 in sequence. 2. Inspect percentage after each save.
Expected Result	All values accepted; percentages are exactly 0, 50 and 100.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-010**
Title	Reject invalid progress without data loss
Related Requirement ID(s)	FR5, NFR5
Preconditions	Alice has b1 page 20 saved.
Steps	1. Attempt -1, 433, 1.5, NaN, Infinity and string "10" through the service. 2. Read the stored page.
Expected Result	Each invalid input is rejected and page 20 remains. UI converts valid numeric input before calling the service.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass

Field	Details
Priority	High
Test Case ID	**TC-011**
Title	Unknown and zero page totals
Related Requirement ID(s)	FR5
Preconditions	b5 has missing pages; Alice has shelved it.
Steps	1. Attempt to save page 0. 2. Repeat with b5 total pages set to zero.
Expected Result	Both saves are rejected as tracking unavailable; percentage is not calculated.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-012**
Title	Prompt before recording completion
Related Requirement ID(s)	FR5, FR4
Preconditions	Alice has b1 Currently Reading.
Steps	1. Save page 432. 2. Inspect status and completion prompt flag. 3. Explicitly move to Completed and repeat save.
Expected Result	Last-page save requests confirmation without changing status; Completed suppresses further prompts.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-013**
Title	Review create edit delete and profile link

Field	Details
Related Requirement ID(s)	FR6
Preconditions	Alice selected, b1 available, no existing review.
Steps	1. Publish 4-star review " Good ". 2. Find it on the book and profile. 3. Edit to 4.25 stars and "Better". 4. Delete the review.
Expected Result	One trimmed review is created; edits replace it; deletion removes it from book and profile.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-014**
Title	Quarter-star boundaries and increments
Related Requirement ID(s)	FR6, FR11
Preconditions	Alice selected and b1 available.
Steps	1. Save every value from 0.25 to 5 in 0.25 increments. 2. Attempt 0, 5.25, 4.1, NaN, Infinity and string "4".
Expected Result	All 20 valid values are accepted; every invalid value is rejected without creating a second review.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-015**
Title	Only edit or delete your own review
Related Requirement ID(s)	FR6, NFR4
Preconditions	Alice has an existing b1 review.

Field	Details
Steps	1. Switch to Bob. 2. Attempt to edit Alice review by its ID. 3. Attempt to delete it. 4. Read the review.
Expected Result	Both operations are rejected and original text remains. This checks demo checks on who owns a record, not security with real sign-in.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-016**
Title	Book-specific threads and replies
Related Requirement ID(s)	FR7
Preconditions	Alice selected; b1 and b2 exist; fixed test clock.
Steps	1. Create a b1 thread. 2. Reply to its ID. 3. Inspect b1 and b2 discussions.
Expected Result	b1 contains thread and reply with author, parent and timestamp; b2 contains neither.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-017**
Title	Reject invalid discussion contributions
Related Requirement ID(s)	FR7
Preconditions	Alice has a thread on b1.
Steps	1. Attempt whitespace-only post. 2. Reply to an absent thread. 3. Reply on b2 using the b1 parent ID.
Expected Result	Every invalid contribution fails; no extra posts are saved.

Field	Details
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-018**
Title	Spoilers hidden until explicit reveal
Related Requirement ID(s)	FR8
Preconditions	Alice posts b1 text "Ending" flagged as spoiler.
Steps	1. Read discussion without revealed IDs. 2. Read it with the post ID explicitly revealed. 3. Read again without that ID.
Expected Result	Hidden view contains no spoiler text; explicit reveal returns text; a new view defaults to hidden.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-019**
Title	Page-based spoiler thresholds
Related Requirement ID(s)	FR8
Preconditions	Future threshold metadata and UI; readers at different progress points.
Steps	1. Flag a post at page 200 of 432. 2. Open as readers below and above page 200. 3. Explicitly reveal as the reader below page 200.
Expected Result	Future threshold feature respects each reader progress and deliberate reveal; no spoiler text leaks into hidden content.
Actual Result	Blocked: US8 threshold extension deferred; current FR8 boolean flag and reveal are tested in TC-018 and TC-039.
Status	Blocked
Priority	Medium

Field	Details
Test Case ID	**TC-020**
Title	EPUB notes privacy and invalid files
Related Requirement ID(s)	FR9, NFR4, NFR8
Preconditions	Future EPUB reader; valid EPUB, corrupt EPUB and non-EPUB files; Alice and Bob.
Steps	1. Open valid EPUB; create, reopen and delete a passage note. 2. Attempt Bob access to Alice note. 3. Import corrupt and unsupported files.
Expected Result	Notes retain correct passage anchors and owner privacy; invalid files show errors without removing saved notes.
Actual Result	Blocked: this feature has not been built yet.
Status	Blocked
Priority	Low
Test Case ID	**TC-021**
Title	Date-filtered reading summaries
Related Requirement ID(s)	FR10, NFR8
Preconditions	Future analytics; known completion dates including date-range boundaries, genres and authors.
Steps	1. Select an inclusive date range. 2. Compare total and per-genre/per-author counts to fixtures. 3. Select an empty range. 4. Attempt Bob access to Alice summary.
Expected Result	Counts match boundary fixtures; empty range is zero; Bob cannot retrieve Alice summary.
Actual Result	Blocked: this feature has not been built yet.
Status	Blocked
Priority	Low
Test Case ID	**TC-022**
Title	Optional sub-ratings warnings and migration

Field	Details
Related Requirement ID(s)	FR11, NFR5
Preconditions	Future enhanced review editor; existing overall-only review.
Steps	1. Reopen old review. 2. Save without sub-ratings. 3. Add prose, characters and plot sub-ratings and warnings. 4. Reject an invalid sub-rating and reopen.
Expected Result	Old review remains readable; fields are optional; valid details persist; invalid save leaves the last saved version intact.
Actual Result	Blocked: this feature has not been built yet.
Status	Blocked
Priority	Medium
Test Case ID	**TC-023**
Title	Buddy-read invitation and membership privacy
Related Requirement ID(s)	FR12, NFR8
Preconditions	Future buddy-read groups; Alice, Bob and non-member Charlie.
Steps	1. Alice creates b1 group and invites Bob. 2. Bob rejects then accepts a new invitation. 3. Charlie attempts direct group read/write. 4. Bob opens discussion after acceptance.
Expected Result	Rejected or unaccepted members have no access; accepted Bob can read/write; Charlie is denied.
Actual Result	Blocked: this feature has not been built yet.
Status	Blocked
Priority	Medium
Test Case ID	**TC-024**
Title	Deterministic community recommendations
Related Requirement ID(s)	FR13

Field	Details
Preconditions	Future recommendation service; books with 3, 2, 2 and 0 reviews.
Steps	1. Request recommendations. 2. Verify descending review counts and alphabetical title tie-break. 3. Follow each book link. 4. Repeat with no reviews.
Expected Result	Correct stable order and detail links; empty activity is handled clearly.
Actual Result	Blocked: this feature has not been built yet.
Status	Blocked
Priority	Medium
Test Case ID	**TC-025**
Title	Verified author application and admin decision
Related Requirement ID(s)	FR14, NFR4
Preconditions	Future author verification; reader, applicant and admin accounts.
Steps	1. Apply for a specific book. 2. Attempt verification as ordinary reader. 3. Approve as admin. 4. Post as the verified author.
Expected Result	Only admin can verify; badge is tied to the approved author/book; posts visibly identify author; failed applications do not grant a badge.
Actual Result	Blocked: this feature has not been built yet.
Status	Blocked
Priority	Medium
Test Case ID	**TC-026**
Title	Follow feed without private activity leakage
Related Requirement ID(s)	FR15, NFR8
Preconditions	Future follow/feed service; Alice and Bob with public reviews and private shelf activity.

Field	Details
Steps	1. Alice follows Bob. 2. Inspect following list, suggestions and feed. 3. Confirm only public activity appears. 4. Unfollow and publish another Bob review.
Expected Result	Following list and eligible review feed update; private progress and shelves never appear; unfollowed future activity is excluded.
Actual Result	Blocked: this feature has not been built yet.
Status	Blocked
Priority	Medium
Test Case ID	**TC-027**
Title	Report and moderate with role enforcement
Related Requirement ID(s)	FR16, NFR4
Preconditions	Future reports and moderation; existing review/thread; reader and admin.
Steps	1. Report content; verify confirmation without automatic removal. 2. Attempt reader access to queue. 3. Admin dismisses one report and removes another item. 4. Resolve a report whose content was already removed.
Expected Result	Reader queue access denied; resolution action, admin and timestamp recorded; removed content hidden; already-removed case handled.
Actual Result	Blocked: this feature has not been built yet.
Status	Blocked
Priority	Medium
Test Case ID	**TC-028**
Title	Measure 100-book browser performance
Related Requirement ID(s)	NFR1
Preconditions	Fresh local browser; 100 generated book fixtures; one selected reader; fixed viewport.

Field	Details
Steps	1. Run 20 title searches. 2. Run 20 alternating shelf-status saves. 3. Time submission through DOM update and two animation frames. 4. Record environment and all timings.
Expected Result	At least 19 of 20 searches AND 19 of 20 shelf updates complete within 2,000 ms.
Actual Result	With 100 books and one reader, all 20 searches and all 20 shelf updates finished within two seconds. The slowest search took 6.5 ms; the slowest update took 6.6 ms.
Status	Pass
Priority	High
Test Case ID	**TC-029**
Title	First-time usability and action count
Related Requirement ID(s)	NFR2
Preconditions	Five first-time participants, fresh demo data and a timer; no coaching.
Steps	1. Give only: "Find Frankenstein, add it, then mark Currently Reading". 2. Count actions from book details to shelf addition. 3. Record completion time and errors for each participant.
Expected Result	Shelf addition takes at most three actions; at least four of five participants finish the full task within three minutes.
Actual Result	Blocked: five first-time participants have not been recruited; no user-study results claimed.
Status	Blocked
Priority	Medium
Test Case ID	**TC-030**
Title	Keyboard access labels and focus sample
Related Requirement ID(s)	NFR3
Preconditions	Browser with Alice selected and fresh interactive page.

Field	Details
Steps	1. Inspect discovery input labels. 2. Focus search once, then type and use Tab/Enter to open Frankenstein. 3. Use keyboard after setting focus to shelf, progress and review controls. 4. Inspect focus outline and textual feedback.
Expected Result	Sample operations work without mouse clicks; discovery controls have labels; focus is visible and status/error text is readable. Full sequential-navigation and screen-reader audit remains required.
Actual Result	Search, open, add, progress and review operated with keyboard after a test script setting the keyboard focus; visible 3px outline and labels checked on discovery. Full manual keyboard and assistive-technology audit pending.
Status	Pass
Priority	Medium
Test Case ID	**TC-031**
Title	Check real sign-in and access rules
Related Requirement ID(s)	NFR4, FR1, FR2, FR4, FR6, FR9
Preconditions	Future server-side sign-in; two accounts; password database and ended session fixtures.
Steps	1. Inspect password hashes. 2. Use Bob credentials to alter Alice profile, shelf, review and note by ID. 3. Sign out Alice and replay the old session.
Expected Result	Every attempt to change another user's data and request using a session that has ended is denied server-side; passwords use a a password-hashing tool designed to store passwords safely.
Actual Result	Blocked: browser demo selection and localStorage cannot meet server-side sign-in or security requirements.
Status	Blocked
Priority	High
Test Case ID	**TC-032**
Title	Keep saved data when the app reopens
Related Requirement ID(s)	NFR5, FR2, FR4, FR5, FR6, FR7

Field	Details
Preconditions	Map-backed test version of browser storage; fresh service.
Steps	1. Save profile, shelf, page, review and discussion. 2. Construct a new service using the same stored JSON. 3. Read each record.
Expected Result	All saved values remain. TC-039 separately verifies an actual browser reload.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-033**
Title	Failed saves keep existing data
Related Requirement ID(s)	NFR5
Preconditions	MemoryStore with one saved shelf; inject save exception "Disk full".
Steps	1. Try changing shelf, progress, removing entry, publishing review/post and editing profile. 2. After each error compare service state and adapter state to original.
Expected Result	Every operation throws; neither last saved state nor in-memory application state changes.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-034**
Title	Change storage without changing the screens
Related Requirement ID(s)	NFR6
Preconditions	MemoryStore and test version of BrowserStore implementing load/save.
Steps	1. Run identical search, shelf and progress assertions with each adapter. 2. Review UI imports and service dependency boundaries.

Field	Details
Expected Result	Same application contract works with both adapters; screens use ReadingService methods; no direct data reads/writes in screen handlers. UI composition still selects an adapter.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	Medium
Test Case ID	**TC-035**
Title	Future test with many books and users
Related Requirement ID(s)	NFR7
Preconditions	Future deployed persistent service; 10,000 books; 50 concurrent authenticated users; recorded hardware.
Steps	1. Run mixed search and shelf-update workload for five minutes. 2. Measure each action from the user's action to the visible result. 3. Calculate fraction within three seconds and error rate.
Expected Result	At least 95% of actions finish within three seconds and error rate is below 1%; publish hardware, workload and raw measurements.
Actual Result	Blocked: persistent multi-user service does not exist; local performance timings cannot validate this target.
Status	Blocked
Priority	Low
Test Case ID	**TC-036**
Title	Private records cannot be retrieved by another account
Related Requirement ID(s)	NFR8, FR4, FR5, FR9, FR10, FR12
Preconditions	Future server with real sign-in; Alice private shelves, progress, EPUB, notes, summary and group.
Steps	1. As Bob request each Alice resource by direct ID and list endpoints. 2. Attempt unaccepted group discussion reads. 3. Verify public profile, review and forum remain accessible.
Expected Result	All access to private records is denied, not just hidden in UI; public resources remain visible.

Field	Details
Actual Result	Blocked: full privacy needs server storage protected by real sign-in; demo isolation in TC-043 is insufficient.
Status	Blocked
Priority	High
Test Case ID	**TC-037**
Title	Render untrusted review text safely
Related Requirement ID(s)	FR6, NFR4
Preconditions	Browser demo with Alice selected and b1 details.
Steps	1. Save review text <img src=x onerror="window.xssExecuted=true">. 2. Inspect rendered DOM and window execution marker.
Expected Result	Text displays literally; no injected image or script executes. This is one rendering check, not a complete security audit.
Actual Result	Review payload rendered as literal text; no image element or script execution.
Status	Pass
Priority	High
Test Case ID	**TC-038**
Title	Guest access boundary
Related Requirement ID(s)	FR1, FR2, FR3, FR4, FR5, FR6, FR7
Preconditions	Fresh service signed out.
Steps	1. Search and read public reviews. 2. Attempt private shelf access and each profile/shelf/progress/review/post mutation.
Expected Result	Public reads work; all listed operations requiring a selected reader are rejected. Real authentication remains deferred.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High

Field	Details
Test Case ID	**TC-039**
Title	Browser reading and discussion journey
Related Requirement ID(s)	FR3, FR4, FR5, FR6, FR7, FR8, NFR5
Preconditions	Fresh browser using the five-book catalogue; Alice and Bob demo identities.
Steps	1. Search Austen and open book. 2. Add, move, reach final page and confirm Completed. 3. Save 4.25-star review; post spoiler thread and reply. 4. Reveal spoiler then switch to Bob. 5. Reload, select Alice and inspect shelf.
Expected Result	Workflow renders correct data; spoiler text absent before reveal and hidden again after reader switch; Bob shelves empty; Alice Completed survives reload; no page errors.
Actual Result	Search, add, move, progress prompt, review, spoiler reveal, reply, reader isolation and real page reload passed; two screenshots saved; no page errors.
Status	Pass
Priority	High
Test Case ID	**TC-040**
Title	Public profile review and discussion visibility
Related Requirement ID(s)	FR2, FR6, FR7, NFR8
Preconditions	Alice selected in fresh service.
Steps	1. Publish review and ordinary post. 2. Sign out and read book reviews, Alice public profile and discussion.
Expected Result	All community-visible content remains readable; no private shelves are included in the public profile object.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-041**

Field	Details
Title	Browser save error and truthful feedback
Related Requirement ID(s)	NFR5
Preconditions	Browser has a saved Completed entry; inject localStorage.setItem failure.
Steps	1. Try changing status to Want to Read. 2. Inspect message. 3. Restore storage and reopen My shelves.
Expected Result	Error text appears instead of success; last saved Completed status remains.
Actual Result	Failed browser save displayed an error and retained Completed; no success message.
Status	Pass
Priority	High
Test Case ID	**TC-042**
Title	Reject unknown resources and unshelved progress
Related Requirement ID(s)	FR3, FR4, FR5, FR6, FR7
Preconditions	Alice selected with no shelves.
Steps	1. Request absent book details. 2. Try shelf, review and thread with absent book ID. 3. Update b1 progress before shelving it.
Expected Result	Every invalid operation is rejected; no shelf records are created.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-043**
Title	Demo reading-list separation
Related Requirement ID(s)	FR4, FR5, NFR8
Preconditions	Alice selected in fresh service.

Field	Details
Steps	1. Alice shelves b1. 2. Switch to Bob and try shelves("alice"). 3. Bob shelves b2. 4. Switch back to Alice.
Expected Result	Bob gets no Alice list and owner override is rejected; Alice still sees only b1. Browser data remains inspectable locally, so this is not full NFR8 checking.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	High
Test Case ID	**TC-044**
Title	Completion date remains stable on repeat saves
Related Requirement ID(s)	FR4
Preconditions	Alice selected; controlled clock can advance one day.
Steps	1. Complete b1. 2. Advance clock and save Completed again. 3. Move away from Completed then complete again.
Expected Result	Repeated same-status save preserves first date; a later new completion records the new date.
Actual Result	Passed. See this test's result in unit-results.tap.
Status	Pass
Priority	Medium

Task Allocation and Timeline

Member	Proposed coding testing and documentation responsibility
Jeffray Zhang	Connect the reading features and prepare the release. Retest shelves and progress, update the test matrix, and collect the submission files.
Liam West	Build real sign-in, logout, and access rules. Test that users cannot change other users' data. Document the setup and security choices.

Member	Proposed coding testing and documentation responsibility
Muskaan Sengupta	Add extra review ratings and content warnings. Check valid and invalid inputs and make sure older reviews still work. Update the FR11 tests.
Tina Kaus	Improve search, shelves, profiles, and custom shelves. Arrange testing with five first-time readers, check keyboard use, and record feedback.
Arya Holmukhe	Improve discussions, spoiler controls, and moderation. Test replies and access rules. Update use cases, problem reports, and sample data for future group tests.

Time window	Deliverables and exit checks
Current checkpoint Nov 7-8	Finish the prototype, 24 unit tests, five browser checks, the 44-case plan, and the report. Check the branches and ITR2.1 tag, then submit the PDF on eClass.
Week 1 Nov 9-15	Build real sign-in and access rules (FR1/NFR4), and start custom shelves. By the end of the week, run sign-in tests and checks for attempts to change another user's data.
Week 2 Nov 16-22	Connect shared storage and the extra review fields. Check reopening, failed saves, and older reviews. Run the five-reader study and full keyboard check. Record results and remaining problems.
Week 3 Nov 23-29	Fix major problems, demonstrate the full workflow, and collect feedback. Check whether the app is ready for the many-user speed test; the single-browser demo cannot prove this.

Code Repository and Branching Strategy

Repository: https://github.com/tqeng/pages-prose

Branch or artifact	Purpose
main	The stable version, updated after tests and report checks pass.
development	The working version where tested feature changes are combined.
feature/core-reading	The main reading and discussion features.
feature/testing	Tests, fixes found during testing, recorded results, and screenshots.

Branch or artifact	Purpose
feature/deliverable-2-1	The test plan, requirement-to-test links, work plan, and submission report.
ITR2.1	The tag for the original stable submission. Check it on GitHub before submitting.
docs/deliverable-2-1.pdf and .md	Formatted eClass submission and editable text source.
docs/testing/	test-cases.json, traceability.csv, test-plan.md, unit-results.tap, browser-results.json and browser-run.txt.
docs/evidence/ and tests/	Actual prototype screenshots and executable unit/browser test suites.

Workflow: Start a feature branch from development. Make one clear change, run its tests, and commit it with a short description. Add the tested change to development. When the full version is ready, check it again and merge it into main. The original submission is marked ITR2.1. Keep the feature branches so the work can be reviewed.